1)

Prove $\forall x, y \in \mathbb{Z}$, If xy is even then x is even or y is even

Contraposition

Let $x, y \in \mathbb{Z}$

Assume x is not even and y is not even

Let $a, b \in \mathbb{Z}$ with $x = 2a+1$

$y = 2b+1$

$xy = 2c+1$

$$xy = (2a+1)y = 2ay + y$$

$$= 2ay + 2b + 1$$

$$2(ay+b)+1$$

$$2c+1$$

xy is not even

$\forall x \in \mathbb{Z}$, x is odd if and only if x is not even

Let $x \in \mathbb{Z}$

Suppose x is odd
and x is even

Choose $a, b \in \mathbb{Z}$, $x = 2a$
$x = 2b + 1$

$$2a = 2b + 1$$

$$2(a-b) = 1$$

$$0 \leq 2(a-b)$$
$$0 \leq a - b$$

$\forall x \in \mathbb{Z}$, $x^2 + x$ is even

1. $\forall x, y, z \in \mathbb{Z}$, If x divides y and y divides z, the $x \mid z$

Let $x, y, z \in \mathbb{Z}$

Assume $x \mid y$ $y \mid z$

Choose $a, b \in \mathbb{Z}$, $y = ax$

$z = bx$

$z = b(ax)$

$\underline{z} = \underline{x(ba)}$

$x \mid z$

$z \% x == 0$

2. $\forall x, y \in \mathbb{Z}$, If xy even and y is odd, then $x+y$ is odd

Let $x, y \in \mathbb{Z}$

Assume xy is even

y is odd

Chose $a, b \in \mathbb{Z}$

$$xy = 2a$$

$$y = 2b+1$$

$$x(2b+1) = 2a$$

$$2xb + x = 2a$$

$$x = 2a - 2bx$$

$$x+y = 2a - 2bx + 2b + 1$$

$$x+y = 2(a - bx + b) + 1$$

$$c = a - bx + b$$

$$2c + 1$$

3.

$\forall x, y \in \mathbb{Z}$, If $x+y$ is odd then $x-y$ is odd

Let $x, y \in \mathbb{Z}$

Assume $x+y$ is odd

$$x+y = 2a+1$$

$$x-y = x+y-2y$$

$$x-y = 2a+1-2y$$

$$x-y = 2a-2y+1$$

$$x-y = 2(a-y)+1$$

$$\underline{2b}+1$$

4. $\forall x, y \in \mathbb{Z}$, If 3 divides xy and $3 \mid y-1$, then $3 \mid x$

Let $x, y \in \mathbb{Z}$

Assume $3 \mid xy$ and $3 \mid y-1$

Choose $a, b \in \mathbb{Z}$

$xy = 3a$ $\quad$ $y-1 = 3b$

$c = a - xb$ $\quad$ $y = 3b+1$

$x(3b+1) = 3a$

$3xb + x = 3a$

$x = 3a - 3xb$

$x = 3(a - xb)$

$x = \underline{3c}$

5. $\forall x \in \mathbb{Z}$, If $2 \mid x$ and $5 \mid x$ then $10 \mid x$

Let $x \in \mathbb{Z}$

Assume $a, b \in \mathbb{Z}$

$x = 2a \qquad x = 5b$

$c = a - 2b$

$x = 5x - 4x$

$x = 5(2a) - 4(5b)$

$x = 10a - 20b$

$10(a - 2b) = 10c$

6. $\forall_{a,x,y} \in \mathbb{Z}$, If $a \mid x+y$ and $a \mid 2x+y$, then $a \mid x$ and $a \mid y$

Let $a, x, y \in \mathbb{Z}$

Assume $a \mid x+y$, $a \mid 2x+y$

Chose $s, t \in \mathbb{Z}$

$x+y = sa \qquad 2x+y = ta$

$u = \qquad\qquad v =$

$x = 2x+y - (x+y) = a(t-s)$

$y = 2x+2y-(2x+y) = 2as - at = a \cdot v$

7. $\forall x \in \mathbb{Z}$, if $\exists a \in \mathbb{Z}, x = 6a+3$

then $\forall n \in \mathbb{N}, \exists b \in \mathbb{Z}$

$x^n = 6b+3$

$x^1 = 6a+3 \ldots\ldots x^n = 6b+3$

$n = 1$

$x = 6a+3$

$x^2 = 36a^2 + 36a + 9$

$= 6(6a^2 + 6a + 1) + 3$

$x^3 = 6 \cdot (\text{int}) + 3$

$x^k = 6(\text{smth}) + 3$

$x^{k+1} = (6a+3)(6k+3)$

15. Let A, B, C be sets. Prove
If $A \subseteq B$, then $A \cup C \subseteq B \cup C$

Assume $A \subseteq B$

Let $x \in A$ or $x \in C$

case 1: $x \in A$

$A \subseteq B$, $x \in B$

$x \in B$ or $x \in C$, which

means $x \in B \cup C$

case 2: $x \in C$

$x \in B$ or $x \in C$

so, $x \in B \cup C$

$A \cup C \subseteq B \cup C$

$$\langle 10 \rangle + \langle 15 \rangle = \langle 5 \rangle$$

$a, b \in \mathbb{Z}$ $\quad x = 10a + 15b$

$$10a + 15b = 5(2a + 3b)$$

$$x \in \langle 5 \rangle$$

let $x \in \langle 5 \rangle$

$u \in \mathbb{Z}$ with $x = 5u$

$v = -u$

$$x = 5u = -10u + 15u$$
$$= 10v + 15v$$